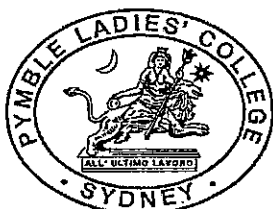


Mr Antonio
Mr Dudley
Mrs Israel
Mrs Kerr
Mrs Soutar

Name:

Teacher:



2010
YEAR 11
PRELIMINARY EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Start each question in a new booklet
- Marks may be deducted for careless or untidy work

Total Marks – 72

- Attempt Questions 1–6
- All questions are of equal value

Mark	/72
Rank	/
Highest mark	/72

Question 1. (12 marks) Use a SEPARATE writing booklet.

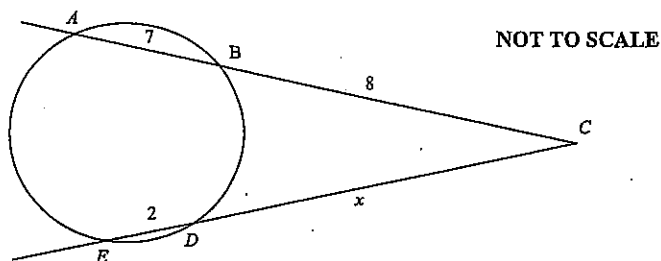
Marks

- (a) Express in simplest form $6 \times 3^n + 3^{n+1}$. 2
- (b) Solve the inequality $\frac{1}{1-x} > 3$. 3
- (c) Evaluate $\lim_{x \rightarrow 2} \frac{2x^2 - 7x + 6}{x - 2}$. 2
- (d) Differentiate $f(x) = 2x + 5$ from first principles. 2
- (e) A curve has parametric equations $x = \frac{t}{2}$ and $y = 3t^2$. 1
Find the Cartesian equation for this curve.
- (f) Find the acute angle between the line $y = 3x + 1$ and $2y = 1 + x$. 2
(Leave your answer correct to the nearest degree.)

Question 2. (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) In the diagram, not drawn to scale, ABC and EDC are straight lines.
 $AB = 7$ cm, $BC = 8$ cm, $DE = 2$ cm. Find CD .



- (b) Find the value(s) of m for which the equation $4x^2 - mx + 9 = 0$ has exactly one real root. 2
- (c) Given that $g(x) = x^4 - kx^3 - 2x + 33$ has a factor of $(x - 3)$, find the value of k . 2
- (d) The point $P(5, -10)$ divides the interval AB , where A is the point $(a, 2b)$ and B point $(3b, -a)$, externally in the ratio $3 : 2$.
 Find the values of a and b . 3
- (e) Find the equation of the locus of a point $P(x, y)$ which moves such that its distance from the point $A(1, 4)$ is four times its distance from the point $B(-2, 1)$. i.e. $PA = 4 PB$. 3

Question 3. (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) Solve the equation $x^2 - 12 = |x|$. 3
- (ii) Sketch the graphs of $y = x^2 - 12$ and $y = |x|$ on a number plane, showing clearly any points of intersection. (Do not find the x -intercepts.) 2
- (iii) Hence solve the inequation $x^2 > |x| + 12$. 1
- (b) (i) Show that $\sin\theta \cos\theta \operatorname{cosec}^2\theta = \cot\theta$. 1
- (ii) Hence solve $\sin^2\theta \cos^2\theta \operatorname{cosec}^4\theta = \frac{1}{4}$ for $0 \leq \theta \leq 360^\circ$. 2
 (Leave your answer correct to the nearest degree.)
- (c) Find the maximum area a triangle can have if the sum of its base and height is 10cm. 3

Question 4. (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the equation of the tangent to the curve $y = x\sqrt{x^2 + 3}$ at the point where $x=1$.

3

- (b) Consider the function $f(x) = \frac{x}{x-1}$.

- (i) Show that $f(x)$ has no stationary points and no inflexion points.

3

- (ii) Sketch the graph, showing asymptotes and intercepts on the axes.

2

Question 4 continues on page 6

Question 4. (continued)

Marks

- (c) RU is a tangent and QT is a diameter of the circle, centre O.

$$\angle SRU = 25^\circ \text{ and } \angle TQS = 30^\circ.$$

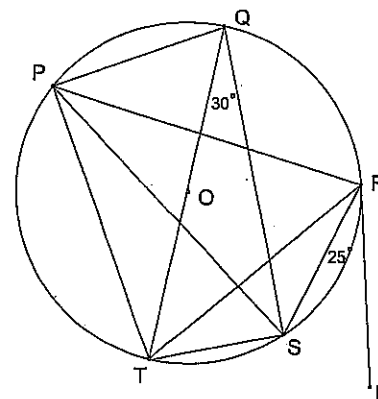
Showing all reasons:

- (i) Prove that $\angle TPR = 55^\circ$.

2

- (ii) Find $\angle QSR$.

2



End of Question 4

Question 5. (12 marks) Use a SEPARATE writing booklet.

Marks

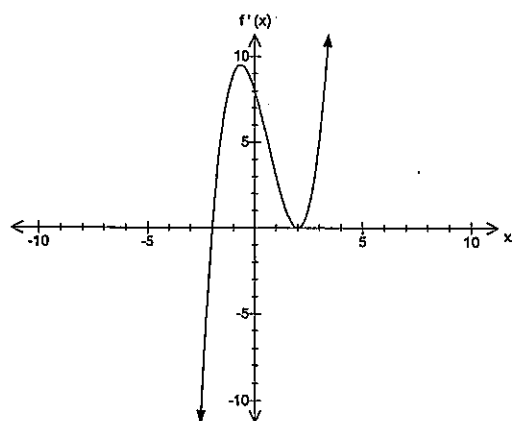
(a) Solve $\left(x + \frac{1}{x}\right)^2 - 5\left(x + \frac{1}{x}\right) + 6 = 0$.

3

(b) The diagram below represents a sketch of $y = f'(x)$.

Given that $f(0) = 10$, in your writing book, sketch neatly a graph of $y = f(x)$.

2



(c) Consider the polynomial $P(x) = x^3 - 2x^2 - 9x - 6$.

(i) Show that $P(-1) = 0$.

1

(ii) Hence express the polynomial as a product of two factors.

2

(iii) Find the exact values of the other zeroes of the polynomial.

1

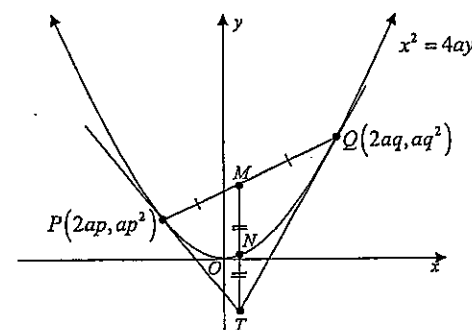
(d) Prove that $\frac{\sin A}{1 + \cos A} = \operatorname{cosec} A (1 - \cos A)$.

3

Question 6. (12 marks) Use a SEPARATE writing booklet.

Marks

(a)



In the diagram, $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are distinct variable points on the parabola $x^2 = 4ay$.

(i) Show that the tangent at P has equation $y = px - ap^2$.

2

(ii) The tangents at P and Q meet at T .

2

Show that T is the point $(a(p+q), apq)$.

(iii) N is the midpoint of MT . Show that as P and Q vary on the parabola $x^2 = 4ay$, N also varies on the parabola $x^2 = 4ay$.

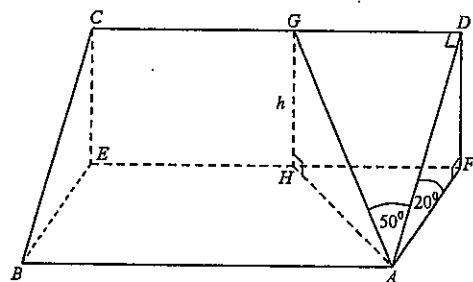
3

Question 6 continues on page 9

Question 6. (continued)

Marks

- (b) A plane hillside $ABCD$ makes an angle of 20° with the horizontal.
 A path AG makes an angle of 50° with a line of greatest slope.
 If $DF = GH = h$:



(i) Show that $AD = \frac{h}{\sin 20^\circ}$. 1

(ii) Show that $AG = \frac{h}{\sin 20^\circ \cos 50^\circ}$. 2

(iii) Hence, find the angle of inclination of the path AG to the horizontal.
 (Leave your answer correct to the nearest degree.) 2

End of paper

Mathematics Ext 1 2010 Yearly

1a) 6×3^n

$$= 6 \times 3^n + 3^n \times 3$$

$$= 3^n (6+3)$$

$$= 3^n \times 9$$

$$= 3^n \times 3^2$$

$$= 3^{n+2}$$

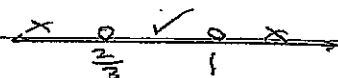
(1 mark)

(1 mark)

b) $\frac{1}{1-x} > 3$

Consider

$$x \neq 1$$



$$\frac{1}{1-x} = 3$$

$$3-3x=1$$

$$-3x = -2$$

$$x = \frac{2}{3}$$

(1 mark)

Test

$$x=0 \quad \frac{1}{1-0} > 3 \text{ False}$$

$$x=\frac{9}{10} \quad \frac{1}{1-\frac{9}{10}} > 3 \text{ True}$$

$$x=2 \quad \frac{1}{1-2} > 3 \text{ False}$$

(1 mark)

$\therefore$ Solution $\{x: \frac{2}{3} < x < 1\}$

(1 mark)

c) $\lim_{x \rightarrow 2} \frac{2x^2 - 7x + 6}{x-2}$

$$= \lim_{x \rightarrow 2} \frac{(2x-3)(x-2)}{x-2} \quad (1 \text{ mark})$$

$$= \lim_{x \rightarrow 2} 2x-3$$

$$= 2 \times 2 - 3$$

$$= 1$$

(1 mark)

(1)

1d) $f(x) = 2x+5$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(x+h)+5 - (2x+5)}{h} \quad (1 \text{ mark})$$

$$= \lim_{h \rightarrow 0} \frac{2x+2h+5-2x-5}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2h}{h}$$

$$= \lim_{h \rightarrow 0} 2$$

$$= 2$$

(1 mark)

e) $xc = \frac{1}{2}$ (1) $y = 3 + x^2$ (2)

From (1) $x = \frac{1}{2y}$ (3)

Sub (3) into (2)

$$y = 3 + \left(\frac{1}{2y}\right)^2$$

$$y = 3 + \frac{1}{4y^2}$$

$$y = \frac{12y^3 + 1}{4y^2}$$

(1 mark)

f) $y = 3x+1$ $2y = 1+x$

$$y = \frac{x}{2} + \frac{1}{2}$$

$$m_1 = 3$$

$$m_2 = \frac{1}{2}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\therefore \tan \theta = 1$$

$$\theta = 45^\circ$$

$$= \left| \frac{3 - \frac{1}{2}}{1 + 3 \times \frac{1}{2}} \right|$$

(1 mark)

$$= \left| \frac{2\frac{1}{2}}{2\frac{1}{2}} \right|$$

$$= 1$$

(1 mark)

(2)

Question 2

a) $5x(x+7) = x(x+2)$ (1 mark)

$$120 = x^2 + 2x$$

$$x^2 + 2x - 120 = 0$$

$$(x-10)(x+12) = 0$$

$$\therefore x = 10 \text{ or } x = -12$$

Since $x = CD$ which is a length, $CD = 10$. (1 mark)

b) For one real root $\Delta = 0$

$$4x^2 - mx + 9 = 0 \quad a=4 \quad b=-m \quad c=9$$

$$\Delta = m^2 - 4 \times 4 \times 9$$

i) $m^2 - 144 = 0$ For one real root. (1 mark)

$$m^2 = 144$$

$$m = \pm 12$$

(1 mark)

c) If $(x-3)$ is a factor of $g(x)$ then $g(3) = 0$

$$\therefore (3)^4 - k(3)^3 - 2(3) + 33 = 0 \quad (1 \text{ mark})$$

$$81 - 27k - 6 + 33 = 0$$

$$108 - 27k = 0$$

$$27k = 108$$

$$k = 4 \quad (1 \text{ mark})$$

(3)

Question 2

2d) $A(a, 2b) \quad B(3b, -a) \quad P(5, -10)$

External

$$\left(\frac{mx_2 + ny_1}{m+n}, \frac{my_2 + nx_1}{m+n} \right) \quad 3: -2$$

m n

$$5 = \frac{-2a + 9b}{1} \quad -10 = \frac{-4b - 3a}{1} \quad (1 \text{ mark})$$

$$5 = -2a + 9b$$

$$-10 = -4b - 3a$$

$$2a - 9b = -5 \quad (1)$$

$$3a + 4b = 10 \quad (2)$$

$$2a - 9b = -5 \quad (1)$$

$$3a + 4b = 10 \quad (2)$$

$$(1) \times 4 \text{ and } (2) \times 9$$

$$8a - 36b = -20 \quad (3)$$

$$27a + 36b = 90 \quad (4)$$

$$(3) + (4)$$

$$35a = 70$$

$$a = 2$$

$$\text{Sub } a=2 \text{ into } (1)$$

$$4 - 9b = -5$$

$$-9b = -9$$

$$b = 1$$

$$\therefore a = 2 \quad b = 1$$

(1 mark for solving)

(4)

2e) $P(x, y) \quad A(1, 4) \quad B(-2, 1)$

$$d_{PA} = \sqrt{(x-1)^2 + (y-4)^2}$$

$$d_{PB} = \sqrt{(x+2)^2 + (y-1)^2}$$

Now $d_{PA} = 4 d_{PB}$

$$\sqrt{(x-1)^2 + (y-4)^2} = 4 \sqrt{(x+2)^2 + (y-1)^2} \quad (1 \text{ mark})$$

$$(x-1)^2 + (y-4)^2 = 16 \{ (x+2)^2 + (y-1)^2 \}$$

$$x^2 - 2x + 1 + y^2 - 8y + 16 = 16 \{ x^2 + 4x + 4 + y^2 - 2y + 1 \}$$

(1 mark) $x^2 - 2x + 1 + y^2 - 8y + 16 = 16x^2 + 64x + 64 + 16y^2 - 32y + 16$

$$15x^2 + 66x + 15y^2 + 24y + 63 = 0$$

$$5x^2 + 5y^2 + 22x - 8y + 21 = 0 \quad (1 \text{ mark})$$

Question 3

a) Consider $x^2 - 12 = 7x$ and $x^2 - 12 = -7x$

$$x^2 - 7x - 12 = 0$$

$$x^2 + 7x - 12 = 0$$

$$(x-4)(x+3) = 0$$

$$(x+4)(x-3) = 0$$

$$\therefore x = 4 \text{ or } x = -3$$

$$\therefore x = -4 \text{ or } x = 3$$

Check solutions

When $x = 4$ LHS = $16 - 12 = 4$ ✓

$x = -4$ LHS = $16 - 12 = 4$ ✓

$x = 3$ LHS = $9 - 12 = -3$ ✗

$x = -3$ LHS = $9 - 12 = -3$ ✗

$\therefore$ Soln $x = 4$ and $x = -4$

a) Consider $x^2 - 12 = 7x$ and $x^2 - 12 = -7x$

$$x^2 - 7x - 12 = 0$$

$$x^2 + 7x - 12 = 0$$

$$(x-4)(x+3) = 0$$

$$(x+4)(x-3) = 0$$

1 mark $\therefore x = 4 \text{ or } x = -3$ (1)

$\therefore x = -4 \text{ or } x = 3$ (1)

Check solutions

When $x = 4$ LHS = $16 - 12 = 4$ ✓

$x = -4$ LHS = $16 - 12 = 4$ ✓

$x = 3$ LHS = $9 - 12 = -3$ ✗

$x = -3$ LHS = $9 - 12 = -3$ ✗

1 mark for

checking

solutions

(1)

$\therefore$ Soln $x = 4$ and $x = -4$

1 mark for solutions

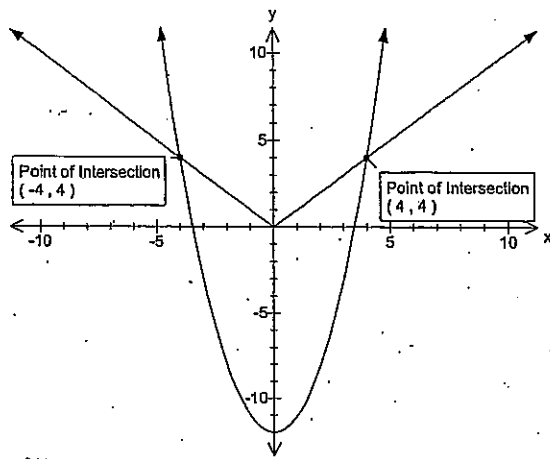
(3)

(5)

(6)

Question 3

a) ii)



1 mark for each graph if all labels, scale and ruled axis are correct.

Max 1 mark if labels or scale missing or if axis not drawn with a ruler.

iii) Soln $\{x: x < -4 \text{ and } x > 4\}$ 1 mark

b) i) $\sin \theta \cos \theta \operatorname{cosec}^2 \theta = \cot \theta$

$$\text{LHS} = \sin \theta \cos \theta \times \frac{1}{\sin^2 \theta}$$

$$= \frac{\cos \theta}{\sin \theta}$$

$$= \cot \theta$$

1 mark

ii) $\sin^2 \theta \cos^2 \theta \operatorname{cosec}^4 \theta = \frac{1}{4}$ Now $\sin \theta \cos \theta \operatorname{cosec}^2 \theta = \cot \theta$

$$\therefore \cot^2 \theta = \frac{1}{4}$$

$$\cot \theta = \pm \frac{1}{2}$$

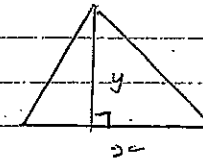
1 mark $\tan \theta = \pm 2$ for $0 \leq \theta < 360^\circ$

1 mark $\theta = 63^\circ 26', 116^\circ 34', 243^\circ 26', 296^\circ 34'$
 $= 63^\circ, 117^\circ, 243^\circ, 297^\circ$

7.

Question 3

c)



base = x height = y

$$A = \frac{1}{2}xy \quad (1)$$

$$x + y = 10 \quad (2)$$

$$y = 10 - x \quad (3)$$

Sub (3) into (1)

$$A = \frac{1}{2}(x)(10 - x)$$

$$A = \frac{1}{2}(10x - x^2) = 5x - \frac{1}{2}x^2 \quad 1 \text{ mark}$$

$$\frac{dA}{dx} = \frac{1}{2}(10 - 2x)$$

$$\frac{dA}{dx} = 5 - x$$

When $\frac{dA}{dx} = 0$ for S.P.

$$5 - x = 0$$

$$x = 5$$

Now $\frac{d^2A}{dx^2} = -1$

< 0 $\therefore x = 5$ will produce a max area. 1 mark

$\therefore$ When $x = 5 \text{ cm}$ $y = 5 \text{ cm}$.

$$\text{Max Area} = \frac{1}{2} \times 5 \times 5$$

$$= \frac{25}{2} \text{ cm}^2 \quad 1 \text{ mark}$$

Question 4

a) $y = x \sqrt{x^2 + 3}$
 $y = x(x^2 + 3)^{\frac{1}{2}}$

$$\frac{dy}{dx} = u \frac{du}{dx} + v \frac{dv}{dx}$$

$$u = x$$

$$\frac{du}{dx} = 1$$

$$v = (x^2 + 3)^{\frac{1}{2}}$$

$$\frac{dv}{dx} = x(x^2 + 3)^{-\frac{1}{2}}$$

(1 mark) $= (x^2 + 3)^{\frac{1}{2}} \cdot 1 + x \cdot x(x^2 + 3)^{-\frac{1}{2}}$

$$= (x^2 + 3)^{\frac{1}{2}} + \frac{x^2}{\sqrt{x^2 + 3}}$$

or

$$\frac{(x^2 + 3) + x^2}{(x^2 + 3)^{\frac{1}{2}}}$$

ISE.

$$= \frac{2x^2 + 3}{(x^2 + 3)^{\frac{1}{2}}}$$

Now when $x=1$ $y=2$

(1 Mark

for for correct gradient and point)

$$y'(1) = \frac{5}{2}$$

Equation of tangent $y - y_1 = m(x - x_1)$

$$y - 2 = \frac{5}{2}(x - 1)$$

$$2y - 4 = 5x - 5$$

(1 mark)

$$5x - 2y - 1 = 0$$

Question 4

b i)

$$f(x) = \frac{x}{x-1}$$

$$f'(x) = \frac{u \frac{du}{dx} - v \frac{dv}{dx}}{v^2}$$

$$u = x$$

$$\frac{du}{dx} = 1$$

$$v = x - 1$$

$$\frac{dv}{dx} = 1$$

$$= \frac{(x-1) \cdot 1 - x \cdot 1}{(x-1)^2}$$

$$= \frac{-1}{(x-1)^2}$$

$$= \frac{-1}{(x-1)^2}$$

$$f''(x) = 2(x-1)^{-3}$$

$$= \frac{2}{(x-1)^3}$$

(1 mark for $f'(x)$ and $f''(x)$ being correct.)

For S.P to occur $f'(x) = 0$

$$= \frac{-1}{(x-1)^2} \neq 0 \quad \text{No S.P (1 mark)}$$

and $f''(x) = \frac{2}{(x-1)^3}$

$$f''(x) \neq 0$$

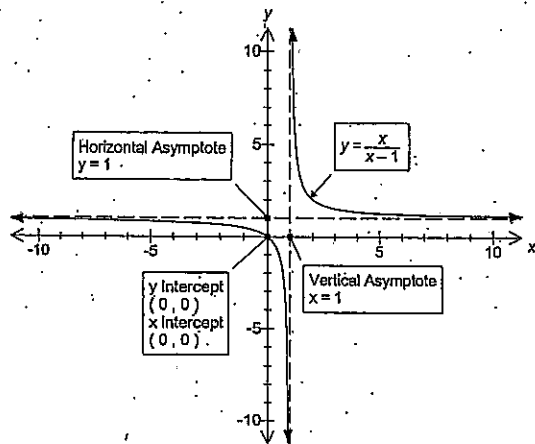
$\therefore$ No point of inflexion (1 mark)

$\therefore$ There are no S.Ps or points of inflexion

Question 4b

11)

2 marks as per graph.



1 mark only if asymptotes, labels, and scale missing.

Must use a ruler.

c) $\angle TPS = 30^\circ$ (Angles at circumference standing on arc TS are equal).

For $\angle SPR = 25^\circ$ (Angle between a tangent and a chord is equal to the angle in the alternate segment).

$$\therefore \angle TPR = 30^\circ + 25^\circ = 55^\circ \quad 1 \text{ mark}$$

1 mark { $\angle TPQ = 90^\circ$ (Angle in a semicircle is a right angle).
 $\angle QPR = 90^\circ - 55^\circ = 35^\circ$

1 mark $\angle QSR = 35^\circ$ (Angles at circumference are equal standing on the same arc QR).

Question 5

a) Let $M = x + \frac{1}{x}$

$$M^2 - 5M + 6 = 0$$

$$(M-3)(M-2) = 0$$

$$\therefore M = 3 \text{ or } M = 2 \quad (1 \text{ mark})$$

$$\therefore x + \frac{1}{x} = 3 \quad \text{or} \quad x + \frac{1}{x} = 2$$

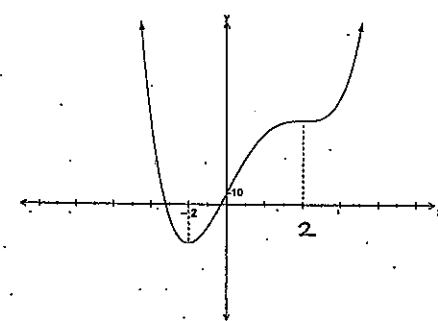
$$x^2 + 1 = 3x \quad \text{or} \quad x^2 + 1 = 2x$$

$$x^2 - 3x + 1 = 0 \quad \text{or} \quad x^2 - 2x + 1 = 0 \quad (1 \text{ mark})$$

$$x = \frac{3 \pm \sqrt{9-4}}{2} \quad \text{or} \quad (x-1)^2 = 0$$

$$\therefore x = \frac{3 \pm \sqrt{5}}{2} \quad \therefore x = 1 \quad (1 \text{ mark for both solutions})$$

b)



2 marks for correct solution.

1 mark if 1 condition below is omitted.

(Maximum 1 mark if poorly labelled, scale missing or not ruled)

The graph passes through (0, 10)

The graph shows a turning point at $x = -2$, anywhere below $y = 10$

The graph shows a point of inflexion at $x = 4$, anywhere above $y = 10$. (11)

Question 5

$$\begin{aligned} \text{c) i) } P(x) &= x^3 - 2x^2 - 9x - 6 \\ P(-1) &= (-1)^3 - 2(-1)^2 - 9(-1) - 6 \\ &= -1 - 2 + 9 - 6 \\ &= 0 \end{aligned} \quad (1 \text{ mark})$$

ii) $\therefore x+1$ is a factor of $P(x)$

$$\begin{array}{r} x^2 - 3x - 6 \\ x+1 \overline{) x^3 - 2x^2 - 9x - 6} \\ \underline{x^3 + x^2} \\ -3x^2 - 9x \\ \underline{-3x^2 - 3x} \\ -6x - 6 \\ \underline{-6x - 6} \\ 0 \end{array} \quad (1 \text{ for division})$$

$$\therefore P(x) = (x+1)(x^2 - 3x - 6) \quad (1 \text{ mark for solution})$$

$$\begin{aligned} \text{iii) } x^2 - 3x - 6 \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{3 \pm \sqrt{9 - 4 \times 1 \times -6}}{2} \\ x &= \frac{3 \pm \sqrt{33}}{2} \quad (1 \text{ for solution}) \end{aligned}$$

$$\text{d) } \frac{\sin A}{1 + \cos A} = \operatorname{cosec} A (1 - \cos A) \quad (\text{Method 1})$$

$$\begin{aligned} \text{LHS} &= \frac{\sin A}{1 + \cos A} \\ &= \frac{\sin A}{1 + \cos A} \times \frac{(1 - \cos A)}{(1 - \cos A)} \quad (1 \text{ mark}) \\ &= \frac{\sin A - \cos A \sin A}{1 - \cos^2 A} \end{aligned}$$

(13)

Question 5

$$\begin{aligned} \text{a) ent'd} \\ &= \frac{\sin A - \sin A \cos A}{\sin^2 A} \\ &= \frac{\sin A (1 - \cos A)}{\sin^2 A} \quad (1 \text{ mark}) \\ &= \frac{1 - \cos A}{\sin A} \\ &= \operatorname{cosec} A (1 - \cos A) \\ &= \text{RHS.} \quad (1 \text{ mark}) \end{aligned}$$

$\therefore \text{LHS} = \text{RHS}$ proven.

Question 6

$$\begin{aligned} \text{a) i) } x^2 &= 4ay \quad \text{Now when } y = 2ap \\ y &= \frac{x^2}{4a} \quad \frac{dy}{dx} = \frac{2ap}{2a} \\ \frac{dy}{dx} &= \frac{2x}{4a} \quad \frac{dy}{dx} = p \\ \frac{dy}{dx} &= \frac{x}{2a} \quad (1 \text{ mark for both differentiations}) \end{aligned}$$

Equation of tangent at P

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - ap^2 &= p(x - 2ap) \\ y - ap^2 &= px - 2ap^2 \end{aligned}$$

$$y = px - ap^2 \quad (1)$$

(1 mark for solution)

(14)

Question 6

ii) Equation of tangent at P
 $y = px - ap^2$ (1)

Equation of tangent at Q
 $y = qx - aq^2$ (2)

(1) = (2)

$px - ap^2 = qx - aq^2$ (1 mark)

$px - qx = ap^2 - aq^2$

$(p-q)x = a(p-q)(p+q)$

$x = a(p+q)$ Sub into (1)

$y = p(a(p+q)) - ap^2$
 $= ap^2 + apq - ap^2$

$y = apq$

(1 mark for solution)

$\therefore T$ is the point $(a(p+q), apq)$

iii) $M(a(p+q), a(\frac{p^2+q^2}{2}))$

$T(a(p+q), apq)$

$\therefore N\left(\frac{2a(p+q)}{2}, \frac{a(p^2+q^2) + apq}{2}\right)$

$\therefore x = a(p+q)$ $y = \frac{a(p^2+q^2)}{2} + \frac{apq}{2}$

(1 mark for x)

Question 6 (ii)

cont'd

$y = \frac{a(p^2+q^2) + 2apq}{4}$

$= \frac{a((p+q)^2 - 2pq) + 2apq}{4}$

$= \frac{a(p+q)^2 - 2apq + 2apq}{4}$

(1 mark for y)

$y = \frac{a(p+q)^2}{4}$ But $x = a(p+q)$

$\therefore p+q = \frac{x}{a}$

$\therefore y = \frac{a}{4} \times \left(\frac{x}{a}\right)^2$

$= \frac{a}{4} \times \frac{x^2}{a^2}$

$y = \frac{x^2}{4a}$

$x^2 = 4ay$ is the locus of N

(1 mark for solution with correct working)

Question 6

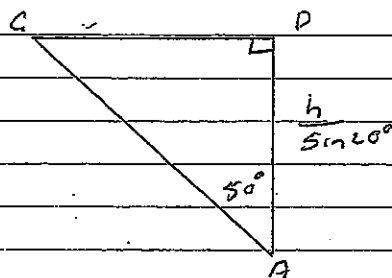
b) i) In $\triangle ADF$

$$\sin 20^\circ = \frac{DF}{AD}$$

$$\sin 20^\circ = \frac{h}{AD}$$

$$AD = \frac{h}{\sin 20^\circ} \quad (1 \text{ mark})$$

ii) In $\triangle ACD$



$$\cos 50^\circ = \frac{h}{\frac{h}{\sin 20^\circ}} \quad (1 \text{ mark})$$

$$AC = \frac{h}{\frac{\sin 20^\circ}{\cos 50^\circ}}$$

$$AC = \frac{h}{\sin 20^\circ \cos 50^\circ} \quad (1 \text{ mark})$$

Question 6

$$\text{iii) } \sin \theta = \frac{h}{AG}$$

$$= \frac{h}{h \sin 20^\circ \cos 50^\circ} \quad (1 \text{ mark})$$

$$\sin \theta = \sin 20^\circ \cos 50^\circ$$

$$\sin \theta = 0.21989431$$

$$\theta = 12^\circ 42'$$

$$= 13^\circ \quad (1 \text{ mark})$$